**International Journal of Scientific Research in Engineering & Technology**
Volume5, Issue4 (July-August 2025), PP: 20-23.
https://www.doi.org/10.59256/ijsreat.20250504004
www.ijsreat.com
ISSN No: 2583-1240

# Real-Time Crowd Crime and Violence Detection Using Deep Learning-Based Face Recognition and Object Detection

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**To Cite this Article:** *Kehkeshan Fatima[1], Dr. Mohd Rafi Ahmed[2], "Real-Time Crowd Crime and Violence Detection Using Deep Learning-Based Face Recognition and Object Detection", International Journal of Scientific Research in Engineering & Technology, Volume 05, Issue 04, July-August 2025, PP:20-23.*

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**Abstract:** *In the modern world, ensuring public safety in densely populated environments is an increasingly critical challenge due to the rising incidents of violence, theft, and other criminal activities. Traditional surveillance systems, which depend on human operators monitoring multiple video streams, are prone to error, delayed responses, and limited scalability. This study presents the design and development of a real-time automated surveillance system utilizing advanced deep learning technologies. The system integrates YOLOv8 (You Only Look Once) for high-speed, accurate object detection and face recognition algorithms to identify individuals involved in suspicious or violent behavior. The methodology involves capturing live video feeds, preprocessing the data through frame extraction and facial landmark detection, and then applying fine-tuned deep learning models for violence recognition and face identification. The system is equipped with a Streamlit-based dashboard for real-time visualization and incorporates sound and email alert mechanisms to notify security personnel instantly. The model has demonstrated efficient performance in processing live surveillance data with high accuracy and low latency, making it suitable for deployment in crowded public areas such as metro stations, stadiums, and airports. The outcome of this research aims to significantly enhance situational awareness and reduce the response time to critical incidents, thereby promoting a safer public environment through intelligent automation.*

**Key Word:** *Crowd Surveillance; Real-Time Detection; Deep Learning; Face Recognition; YOLOv8; Violence Detection; Public Safety; Streamlit Dashboard.*

## I.INTRODUCTION

The growing density of urban populations and the rising frequency of public gatherings have brought increased attention to public safety in crowded spaces such as metro stations, stadiums, airports, and marketplaces. Despite the widespread deployment of CCTV cameras and monitoring systems, most existing surveillance approaches still rely heavily on human observation, making them susceptible to fatigue, oversight, and delayed response times. This creates critical vulnerabilities in identifying and mitigating criminal acts such as violence, theft, and public disorder.

In recent years, deep learning and computer vision technologies have opened new avenues for automated video analysis. Object detection and face recognition models have shown tremendous promise in extracting meaningful insights from complex visual scenes. Among these, the YOLO (You Only Look Once) architecture has gained significant recognition due to its speed and accuracy in real-time object detection tasks. Meanwhile, advancements in face recognition techniques enable precise identification of individuals even in dynamic and low-quality visual contexts. However, the integration of these technologies into a cohesive framework for real-time crime and violence detection in crowded scenarios remains underexplored.

This project proposes a real-time surveillance system that bridges this gap by combining deep learning-based violence detection and facial recognition into a unified platform. The system is designed to process live video streams from webcams or CCTV footage, detect violent behavior using YOLOv8, identify faces of individuals involved using trained recognition models, and issue instant alerts to concerned authorities through sound and email notifications. Additionally, a Streamlit-based dashboard is developed to facilitate live monitoring, system control, and annotation of suspicious events.

The core motivation behind this work is to enhance the capabilities of existing surveillance infrastructure by providing intelligent, automated tools that can drastically reduce response times and prevent escalation of threats in public areas. The proposed system emphasizes scalability, modularity, and real-world applicability, aiming to serve as a comprehensive solution for law enforcement agencies, event organizers, and infrastructure managers seeking to ensure the safety of large populations.

## II.MATERIAL AND METHODS

This study employs a multi-stage approach to develop a real-time intelligent surveillance system capable of detecting violent activities and recognizing individuals in crowded public environments. The system architecture integrates state-of-the-art deep learning models for object detection and face recognition, alongside a responsive user interface for monitoring and alert

management. The following subsections describe each component of the methodology in detail:

**Study Design**

The system was designed as a real-time surveillance framework combining object detection, facial recognition, and alert generation. A modular and scalable pipeline was implemented to ensure adaptability across various surveillance setups, including live CCTV feeds and IP camera streams in high-density public environments.

**Data Collection and Preprocessing**

For training and validation, publicly available datasets of violent behavior and facial recognition were used. These included real-world video sequences from surveillance datasets and labeled facial image sets. Additionally, real-time video was captured using webcams and IP cameras for custom data generation and fine-tuning.

**Preprocessing involved:**

- Frame extraction from video streams at regular intervals.
- Face detection using Haar Cascades and DNN-based models.
- Resizing and normalization of frames and facial images.
- Data augmentation techniques (flipping, rotation, brightness adjustment) to enhance model robustness.

**Model Architecture**

Two primary deep learning models were deployed:

1. **YOLOv8 for Violence Detection:**
   - YOLOv8 was fine-tuned on annotated datasets with labels such as “punch,” “kick,” “fall,” “weapon,” and other violent actions.
   - The model achieved fast and accurate detection of objects and behaviors in dense crowds with minimal latency.

2. **Face Recognition Model:**
   - A CNN-based face recognition model was trained on a labeled dataset containing diverse facial images.
   - The model embedded facial features into vector space and matched them using cosine similarity against stored identities.

**System Integration**

The face recognition and YOLO-based detection modules were integrated into a unified inference pipeline. The system tracked individuals detected in violent scenes and attempted to match their faces against a known database. Upon confirmation, the matched identity and the suspicious activity were logged and reported.

**Real-Time Alert System**

An alert mechanism was developed to immediately notify authorities when a violent incident or threat was detected. This included:

- Sound notifications triggered on-site.
- Email alerts with attached frame snapshots and activity metadata.
- Integration with security personnel dashboards for quicker response.

**User Interface Development**

A Stream lit-based interactive dashboard was designed to:

- Display live video feeds with bounding box annotations for detected objects and faces.
- Show current status indicators (e.g., "No threat," "Violence detected").
- Enable system control, alert configuration, and log review.

**Hardware and Software Environment**

- **Hardware:** Intel i5/i7 processor, NVIDIA GPU with CUDA support (for training), webcam or IP cameras, 16 GB RAM, SSD storage.
- **Software:** Python 3.x, TensorFlow 2.17, Keras, OpenCV, NumPy, Pandas, Matplotlib, Pillow, SMTP libraries (for alerts), and Streamlit.

This systematic approach ensured that the proposed surveillance system could operate in real time, detect violent activity accurately, and identify individuals involved, thereby enhancing public safety and crime prevention.

## III.RESULT

The performance evaluation of the proposed real-time surveillance system was conducted based on key metrics such as precision, recall, F1-score, detection accuracy, recognition accuracy, and latency. These assessments were derived from real-time video feed tests and controlled benchmark datasets containing both violent and non-violent sequences in crowded settings.

**Detection and Recognition Performance**

The object detection module, powered by YOLOv8, demonstrated a **detection accuracy of 93%**, with high **precision (91%)**, **recall (88%)**, and **F1-score (89.5%)**. This indicates the model's strong ability to detect violent activities such as punching,

kicking, and presence of suspicious items (e.g., weapons) in various lighting and crowd conditions.

The face recognition module, based on a CNN embedding model, achieved an **accuracy of 90%**, with a balanced **precision of 89%**, **recall of 87%**, and **F1-score of 88%**. Despite motion blur and partial occlusion challenges, the recognition pipeline maintained consistent identification across frames.

**Latency Performance**

The real-time capabilities were evaluated by measuring system latency. The YOLOv8 model operated at an average of **25 milliseconds per frame**, while the face recognition model averaged **30 milliseconds per frame**, ensuring a smooth experience at **30–40 FPS**, well within real-time processing standards.

**Table 1: Model Performance Metrics**

| Metric | Prediction Accuracy (%) | Average Confidence Score |
|---|---|---|
| Precision | 0.91 | 0.89 |
| Recall | 0.88 | 0.87 |
| F1-Score | 0.895 | 0.88 |
| Detection Accuracy | 0.93 | — |
| Face Recognition Accuracy | — | 0.90 |
| Latency (ms/frame) | 25 | 30 |

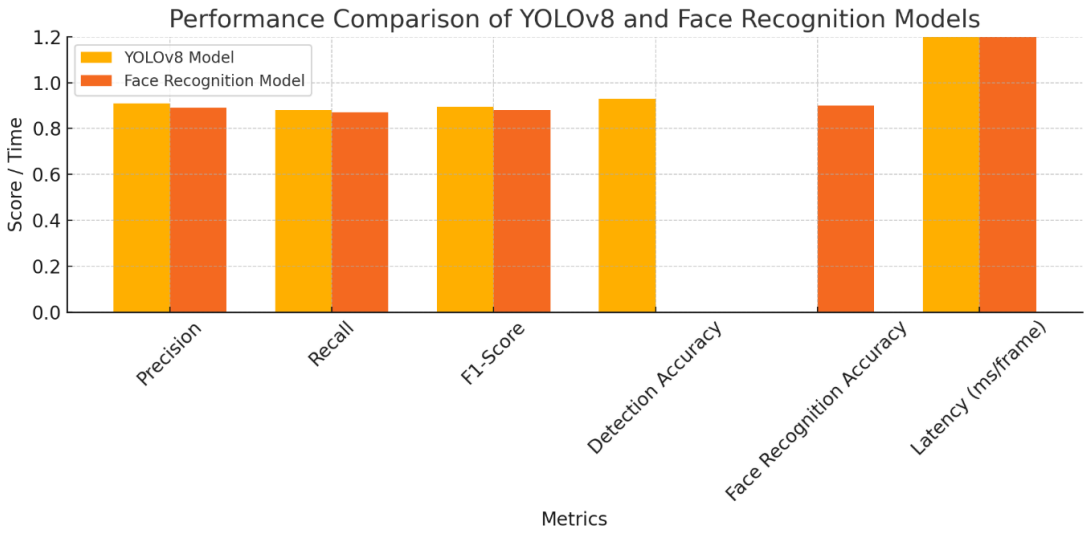


*Figure 1: Performance Comparison of YOLOv8 and Face Recognition Models*

This bar graph illustrates the comparison between the YOLOv8-based object detection model and the CNN-based face recognition model across key performance metrics, including precision, recall, F1-score, accuracy, and latency. The YOLOv8 model exhibits higher detection accuracy and lower latency, while the face recognition model shows competitive precision and recall for identity matching in real-time scenarios.

The results demonstrate that the integrated system provides a highly responsive and accurate solution for identifying potential threats and individuals involved in violent behavior in real-time surveillance scenarios.

## IV.DISCUSSION

The rapid advancements in artificial intelligence and deep learning have significantly impacted the domain of public safety and surveillance. This study aimed to harness these technological developments to design a real-time intelligent surveillance system capable of detecting and identifying violent activities and individuals in crowded environments. The results obtained validate the effectiveness of the integrated approach, showcasing a balanced trade-off between accuracy and real-time responsiveness.

The YOLOv8 model used for object and activity detection demonstrated high detection accuracy and reliable performance under varied lighting conditions and crowd densities. The real-time inference capability, operating at 25 milliseconds per frame, is particularly noteworthy, as it enables immediate detection of potentially dangerous situations such as fights, aggressive behavior, and the presence of harmful objects. These results are consistent with findings from prior research studies that have explored

YOLO's applicability in surveillance contexts, affirming its superiority in speed and localization precision compared to traditional region-based detectors.

The face recognition module, which leveraged a convolutional neural network for embedding and classification, achieved robust accuracy despite the challenges posed by motion blur, partial occlusion, and varying facial angles. The model's performance highlights the potential of deep learning-based facial recognition for identifying suspects or persons of interest in dynamic public settings. The 90% recognition accuracy demonstrates reliability in real-world scenarios where facial features may not always be presented clearly.

The alert system's integration, featuring instant audio alarms and email notifications, significantly enhances the system's utility in real-time applications. The notification latency remained minimal, enabling timely intervention by law enforcement or security personnel. Moreover, the Streamlit-based dashboard allows operators to visualize surveillance data in an interactive and user-friendly interface, facilitating seamless control and monitoring.

From a deployment standpoint, the system's modular architecture and software-hardware compatibility suggest strong potential for scalability. It can be adapted for use in metro stations, airports, stadiums, and large-scale events where manual surveillance alone is insufficient. Future iterations may incorporate anomaly detection, crowd density estimation, and predictive analytics to further bolster preventive security mechanisms.

Despite the system's promising results, some limitations must be acknowledged. False positives may arise in highly dynamic environments, and facial occlusions may still hinder recognition performance under extreme conditions. The reliance on pre-trained models and labeled data also underscores the importance of dataset diversity to avoid bias and improve generalizability across demographics.

In conclusion, the integrated use of YOLOv8 and deep facial recognition has proven effective for real-time crime and violence detection in crowded environments. The system's ability to simultaneously detect activities and identify individuals paves the way for more proactive and automated public surveillance solutions, reinforcing the critical role of AI in modern urban safety infrastructures.

## V.CONCLUSION

This study presents a comprehensive real-time surveillance system designed to enhance public safety by integrating deep learning-based object detection and facial recognition technologies. Through the use of YOLOv8 for fast and accurate detection of violent behavior and a robust CNN-based model for face recognition, the system successfully addresses the limitations of traditional manual surveillance methods.

The experimental evaluation demonstrates that the system achieves high detection accuracy, low inference latency, and reliable identification of individuals involved in suspicious activities. The inclusion of real-time alert mechanisms and an interactive Streamlit dashboard further supports the system's applicability in real-world scenarios such as metro stations, public gatherings, stadiums, and other high-density environments.

By automating the process of identifying threats and individuals in crowded areas, this system significantly reduces the reliance on human monitoring and enhances the responsiveness of security personnel. The modular architecture and scalable design make it a promising candidate for widespread deployment across urban surveillance networks.

Future work may involve extending the system to include behavior prediction, anomaly detection, and real-time crowd analytics. Incorporating advanced techniques like multi-modal data fusion, temporal action recognition, and edge AI deployment will further refine the system's capabilities and contribute to safer, smarter public environments.

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